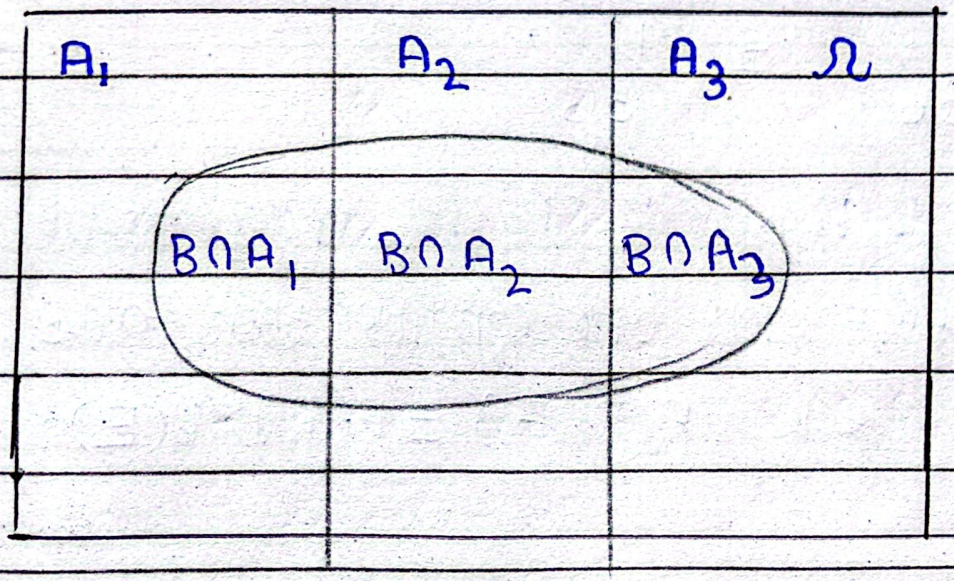
Section : 4.7
Baye's Rule

Imp Point :-

If two events are both exhaustive and mutually excusive exactly one of them must happen.



The Rectangle Represent all possible Students. Divide into three groups A_1, A_2, A_3.

Two Thing we can notice in Figure:

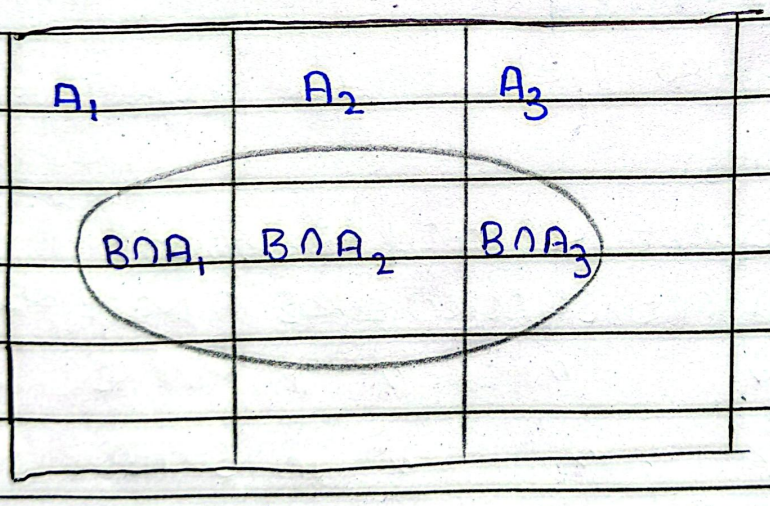
1. ALL three sections cover entire rectangle so they are exhaustive.

2. None of the section overLap so they are mutually exclusive.

Now whats B in Figure:

Prob that we want to Find randomLy selected student who passed an exam.

B: Event (pass the exam)



- The oval Represent event B.
- The oval cut across all three groups.
- This create three smaller regions.

- $A_1 \cap B$: Computer Science Std who passed
- $A_2 \cap B$: Business Std who passed
- $A_3 \cap B$: Engineering Std who passed.

Date: _____ Day: _____

Rule of Total Probability:

Suppose we want to calculate the prob that any randomly selected who passes

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + P(A_3 \cap B)$$

applying general multiplication rule.

$$P(B) = P(A_1) \cdot P(B|A_1) + P(A_2) \cdot P(B|A_2) + P(A_3) \cdot P(B|A_3)$$

↓

and this Formula is known an rule of total probability.

Example: 4.24

as Pg 190

Let events:

S : resident selected is senior

R_1 : resident selected Lives in Northeash

R_2 : " " " in Midwest

R_3 : " " " in South

R_4 : " " " in West

Given:

$P(R_1) = 18.3\% = 0.183$	$P(S\|R_1) = 13.6\%$
$P(R_2) = 22.2\% = 0.222$	$P(S\|R_2) = 12.8\%$
$P(R_3) = 36.3\% = 0.363$	$P(S\|R_3) = 12.5\%$
$P(R_4) = 23.2\% = 0.232$	$P(S\|R_4) = 11.2\%$

Find:

Percentag of US residents that are seniors.

$$\therefore P(S) = P(R_1)\cdot P(S|R_1) + P(R_2)\cdot P(S|R_2) + P(R_3)\cdot P(S|R_3) + P(R_4)\cdot P(S|R_4)$$

$$P(S) = (0.183)(0.136) + (0.222)(0.128) + (0.363)(0.125) + (0.232)(0.112)$$

$$P(S) = 0.1247$$

$P(S) = 12.47\%$ of US residents are seniors.

Baye's Rule:-

As we have Discussed Total prob rule now we can easily understand baye's rule.

Suppose A_1 and A_2 A_3 are mutually exclusive & Exhaustive and [redacted] we want to Find $P(A_2|B)$

Using Conditional Prob.

$$P(A_2|B) = \frac{P(A_2 \cap B)}{P(B)}$$

Using multiplication rule

$$P(A_2 \cap B) = P(A_2) \times P(B|A_2)$$

So

$$P(A_2|B) = \frac{P(A_2)\,P(B|A_2)}{P(B)}$$

Date: Day:

Using Total Prob Rule.

$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)$$

Final Bayes' Rule.

$$P(A_2|B) = \frac{P(A_2)(B|A_2)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)}$$

Prior Probs.

similarly we can Find $P(A_1|B)$ and $P(A_3|B)$ by replacing A_2 in numerator with A_1 and A_3.

When to Use Conditional Prob and Baye's Rule.

Conditional: used when know Joint & Marginal Prob.

Bayes Rule: Used when we know reverse conditional probs and prior probs.

Example: 4.25. (use event & Data from 4.24

$P(R_1) = 18.3\%$
$P(R_2) = 22.2\%$
$P(R_3) = 36.3\%$
$P(R_4) = 23.2\%$

$P(S|R_1) = 13.6\%$
$P(S|R_2) = 12.8\%$
$P(S|R_3) = 12.5\%$
$P(S|R_4) = 11.2\%$

Date: Day:

Find:-

What percentage of seniors are northeast residents?

$$P(R_1|S) = \frac{P(R_1)P(S|R_1)}{P(R_1)P(S|R_1)+P(R_2)P(S|R_2)+P(R_3)P(S|R_3)+P(R_4)P(S|R_4)}$$

$$P(R_1|S) = \frac{(0.183)(0.136)}{(0.183)(0.136)+(0.222)(0.128)+(0.363)(0.125)+(0.232)(0.112)}$$

$$P(R_1|S) = \frac{0.0249}{0.1247} = 0.1997$$

$P(R_1|S) = 19.97\%$ of seniors are Northeast residents.

Example: 4.26

Let events:

S = selected person is a smoker.

L_1 = Person selected has Lung Disease

L_2 = Person selected has no Lung Disease

Given Data:

$P(L_1) = 7\% = 0.07$

$P(S|L_1) = 90\% = 0.9$

$P(S|L_2) = 25.3\% = 0.253$

$P(L_2) = 1 - 0.7 = 0.93$

Date: _____ Day: _____

Find:- Smoker

Find Prob that selected has Lung Disease : $P(L_1 | S)$

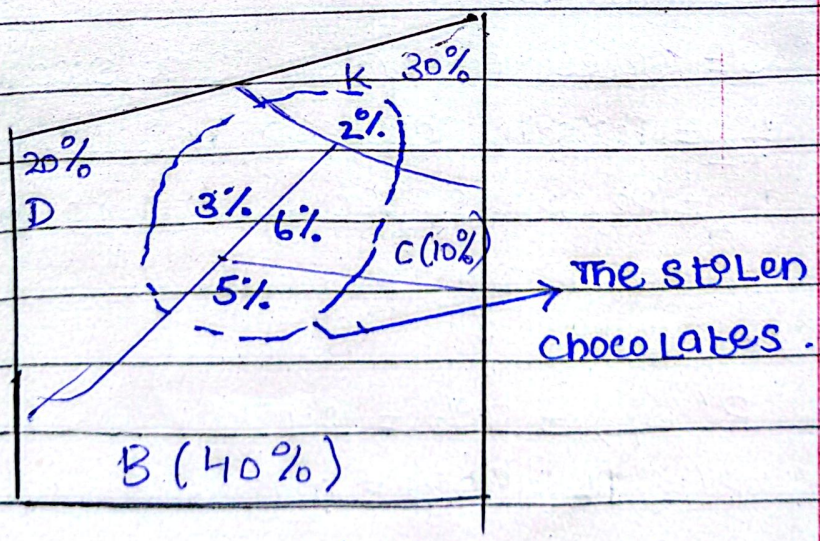
$$P(L_1|S) = \frac{P(L_1) \cdot P(S|L_1)}{P(L_1) \cdot P(S|L_1) + P(L_2) P(S|L_2)}$$

$$P(L_1|S) = \frac{(0.07)(0.9)}{(0.07)(0.9) + (0.93)(0.253)}$$

$$P(L_1|S) = \frac{0.063}{0.063 + 0.23529} = 0.2112$$

$P(L_1|S) = 21.12\%$ of smokers have Lung Disease

Example: (class)



Dairy Milk :→ 20% → D

kitkat → 30% → K

Bubbly → 40% → B

coconet → 10% → C

Let Event S: Stolen.

$P(S|D) = 3\%$ $P(S|K) = 2\%$

$P(S|B) = 5\%$ $P(S|C) = 6\%$

Given that chocolate has been stolen

Find prob that it was bubbly $P(B|S)$

$$P(B|S) = \frac{P(B)\,P(S|B)}{P(D)P(S|D) + P(K)P(S|K) + P(B)P(S|B) + P(C)P(S|C)}$$

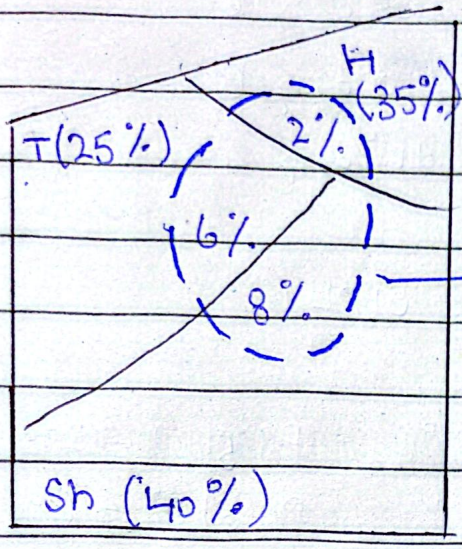
$$P(B|S) = \frac{(0.4)(0.05)}{(0.2)(0.03) + (0.3)(0.02) + (0.4)(0.05) + (0.1)(0.06)}$$

$$P(B|S) = \frac{0.020}{0.038} = 0.5263$$

$P(B|S) = 52.63\%$

Date: _____ Day: _____

Example: (class)



Taimoor → 25% → T
Hamza → 35% → H
Shahab → 40% → Sh

Let event R: Replied to an unsaved no.

$P(R|T) = 6\%$ $P(R|H) = 2\%$
$P(R|Sh) = 8\%$

Given that Shahab replied to an unsaved number, what prob that reply went to Taimoor.

$$P(T|R) = \frac{P(T)\,P(R|T)}{P(T)(R|T) + P(H)\,P(R|H) + P(Sh)\,P(R|Sh)}$$

Date: Day:

$$P(T|R) = \frac{(0.25)(0.06)}{(0.25)(0.06) + (0.35)(0.02) + (0.40)(0.08)}$$

$$P(T|R) = \frac{0.015}{0.054} = 0.2778$$

$$P(T|R) = 27.78\%$$

Exercise: 4.7

Question: 4.153

Qs Pg 194

a.

Four events are exhaustive if they cover all possible outcomes. At least one of four must happen.

Example: Rolling a Die

- A: get 1 or 2
- B: getting 3
- C: getting 4 or 5
- D: getting 6

Together A, B, C and D cover all possible outcome.

Date:_____ Day:_____

b-

Four events are mutually exclusive if they can't happen at same time. This means if one event happen others can't happen.

Example:-
when rolling a Die, getting 1, 2, 3, 4, 5 and 6 are mutually exclusive.

c-

No, Exhaustive events cover all possible outcomes but they can overlap.

Example: Student in a class.
Suppose class has 4 students, Ali, Ahmed Sara and Haroon.

A: Std who play cricket
B: Std who play football
C: Std who play hockey
D: Std who play tennis

These are exhaustive events coz every Std is included in atleast one event. But they are not mutually exclusive coz some std play more that one sport.

Date: _____ Day: _____

d-

No, Mutually exclusive events can't happen together. and they don't necessarily covers all possible outcomes.

Example: Rolling a Die.

A = getting 1

B = getting 2

C = getting 3

D = getting 4

These 4 events are mutually exclusive but not exhaustive.

Question: 4.155

Qs Pg 194 and Data Pg 190.

a-

Let event R_3: Southern Residents.

$P(R_3) = 36.3\%$

b-

Let event S: Senior Citizen

Let even R_3: Southern Resident.

$P(S|R_3) = 12.5\%$

Date: _____ Day: _____

C-

Let event R_3 : Souther Residents

Let S : Seniors.

$P(R_3 | S) = ?$

$$P(R_3|S) = \frac{P(R_3)\,P(S|R_3)}{P(R_1)P(S|R_1) + P(R_2)P(S|R_2) + P(R_3)P(S|R_3) + P(R_4)P(S|R_4)}$$

$$P(R_3|S) = \frac{(0.363)(0.125)}{(0.183)(0.136) + (0.222)(0.128) + (0.363)(0.125) + (0.232)(0.112)}$$

$$P(R_3|S) = \frac{0.045375}{0.124663} = 0.364$$

$P(R_3|S) = 36.4\%$.

Question: 4.157

as Pg 194.

Let events:

M: adults who are male

F: adults who are Female

A: A

Date: _____ Day: _____

Given Data:

$P(M) = 48\% = 0.48$

$P(F) = 52\% = 0.52$

$P(A|M) = 54\% = 0.54$

$P(A|F) = 33\% = 0.33$

a:

$P(A) = ?$

$\therefore$

$P(A) = P(M)\,P(A|M) + P(F)\,P(A|F)$

$P(A) = (0.48)(0.54) + (0.52)(0.33)$

$P(A) = 0.2592 + 0.1716$

$P(A) = 0.4308$

$P(A) = 43.08\%$

b-

$P(A|F) = ?$

$P(A|F) = 33\%$

c-

$P(F|A) = ?$

$$P(F|A) = \frac{P(F)\,P(A|F)}{P(M)\,P(A|M) + P(F)\,P(A|F)}$$

$$P(F|A) = \frac{(0.52)(0.33)}{(0.48)(0.54) + (0.52)(0.33)}$$

Date: Day:

$$P(F|A) = \frac{0.1716}{0.4308} =$$

$P(F|A) = 0.3983$

$P(F|A) = 39.83\%$

Question: 4.159

Qs Pg 194

Let events:-

R: adults who read astrology report

L: Adult having educational LVL Less than High school

H: Adult having educational LVL High school graduate

B: Adult is educational level is Baccalaureate or higher.

Given Data:

$P(L) = 7.4\% = 0.074$

$P(H) = 53.3\% = 0.533$

$P(B) = 39.3\% = 0.393$

$P(R|L) = 9\% = 0.09$

$P(R|H) = 7\% = 0.07$

$P(R|B) = 4\% = 0.04$

Date: _____ Day: _____

a-

$P(R) = ?$

$\therefore P(R) = P(L)\,P(R|L) + P(H)\,P(R|H) + P(B)\,P(R|B)$

$P(R) = (0.074)(0.09) + (0.533)(0.07) + (0.393)(0.04)$

$P(R) = 0.00666 + 0.03731 + 0.01572$

$P(R) = 0.05969$

$P(R) = 5.969\,\%$

b-

Is not a high school graduate given he or she reads astrology report

↓

This mean adult educational Level is than high school graduate

↓

$P(L|R)$

$\therefore P(L|R) = \dfrac{P(L)\,P(R|L)}{P(L)\,P(R|L) + P(H)\,P(R|H) + P(B)\,P(R|B)}$

$P(L|R) = \dfrac{(0.074)(0.09)}{(0.074)(0.09) + (0.533)(0.07) + (0.393)(0.04)}$

$P(L|R) = \dfrac{0.00666}{0.05969} = 0.1116$

$P(L|R) = 11.16\,\%$

Date: _____ Day: _____

C-

$P(B|R)$

$$\therefore P(B|R) = \frac{P(B)P(R|B)}{P(L)P(R|L)+P(H)P(R|H)+P(B)P(R|B)}$$

$$P(B|R) = \frac{(0.393)(0.04)}{0.05969} = \frac{0.01572}{0.05969} = 0.2634$$

$P(B|R) = 26.34\%$

Question: 4.161

Qs Pg 194-195.

Let events:-

A_1: Adult b/w 18-34

A_2: Adult b/w 35-49

A_3: Adult b/w 50-64

A_4: 65 & over.

O: Adult is overweight or obese.

Given Data:

$P(A_1) = 42.5\% = 0.425$

$P(A_2) = 28.5\% = 0.285$

$P(A_3) = 16.4\% = 0.164$

$P(A_4) = 12.6\% = 0.126$

Date: _____ Day: _____

$P(O|A_1) = 41.1\% = 0.411$

$P(O|A_2) = 57.9\% = 0.579$

$P(O|A_3) = 68.2\% = 0.682$

$P(O|A_4) = 55.3\% = 0.553$

a-

$P(O) = ?$

$P(O) = P(A_1)P(O|A_1) + P(A_2)P(O|A_2) + P(A_3)P(O|A_3) + P(A_4)P(O|A_4)$

$P(O) = (0.425)(0.411) + (0.285)(0.579) + (0.164)(0.682) + (0.126)(0.553)$

$P(O) = 0.174675 + 0.165015 + 0.111848 + 0.069678$

$P(O) = 0.521216$

$P(O) = 52.12\%$

b-

$P(O|A_2) = ?$

$P(O|A_2) = 57.9\%$

Date: _____ Day: _____

c-

$P(A_2|O) = ?$

$$P(A_2|O) = \frac{P(A_2)P(O|A_2)}{P(A_1)P(O|A_1) + P(A_2)P(O|A_2) + P(A_3)P(O|A_3) + P(A_4)P(O|A_4)}$$

$$P(A_2|O) = \frac{(0.285)(0.579)}{0.521216} = \frac{0.165015}{0.521216} = 0.31656$$

$P(A_2|O) = 31.66\ \%$

d-

For Part (a)

Approx 52.12% of all Utah adults are overweight or obese.

For Part (b)

Among Utah adults aged 35-49, approx 57.9% are overweigh or obses.

For Part (c):

Among all Utah adults who are overweigh or obese approx 31.66 % are age 35-49.

Date: _____ Day: _____

Question : 163

QS Pg 195

Let events.

M = Books sells more than Projected

C = close to Projected

L = Less than Projected

R = Revised For second edition.

Given data:

$P(M) = 10\% = 0.10$

$P(C) = 30\% = 030$

$P(L) = 60\% = 0.60$

$P(R|M) = 70\% = 0.70$

$P(R|C) = 50\% = 0.50$

$P(R|L) = 20\% = 0.20$

Q:

$P(R) = ?$

$P(R) = P(M)\,P(R|M) + P(C)(P(R|C) + P(L)\,P(R|L)$

$P(R) = (0.10)(0.70) + (0.30)(0.50) + (0.60)(0.20)$

$P(R) = 0.07 + 0.15 + 0.12 = 0.34$

$P(R) = 34\%$

Date: _____ Day: _____

b-

$P(L|R) = ?$

$$P(L|R) = \frac{P(L)P(R|L)}{P(M)P(R|M) + P(C)P(R|C) + P(L)P(R|L)}$$

$$P(L|R) = \frac{(0.60)(0.20)}{0.34} = \frac{0.13}{0.34} = 0.35294$$

$P(L|R) = 35.294\ \%$

Question : 4.165

Let events:-

D = Person has Disease

D' = Person does't have Disease

T = Test result is tive

T' = Test result is not positive\negative.

Given:

Sensitivity:-

Prob that person having Disease will test Positive.

$P(T|D) = 0.934$

Date: Day:

Specificity:

Prob that a person not having Disease will test negative.

$P(T'|D') = 0.968.$

Its also known Roughly 1 in 500 People has Disease

$P(D) = \frac{1}{500} = 0.002$

$P(D') = 1 - 0.002$

$P(D') = 0.998.$

a-

sensitivity: 93.4% of people who actually have Disease will test positive.

Specificity: 96.8% of people who don't have disease will test negative.

b-

$P(D|T)$:

$P(D|T) =$

$$P(D|T) = \frac{P(D)\,P(T|D)}{P(D)\,P(T|D) + P(D')\,P(T|D')}$$

Date: Day:

we need to Find this term P(T|D') First:

↓

Prob of +ive test when person does't have Disease

$$P(T|D') = 1 - P(T'|D')$$
$$P(T|D') = 1 - 0.968$$
$$P(T|D') = 0.032$$

$$P(D|T) = \frac{(0.002)(0.934)}{(0.002)(0.934) + (0.998)(0.032)}$$

$$P(D|T) = \frac{0.001868}{0.033804} = 0.05526$$

$$P(D|T) = 5.526\ \%$$

C-

5.526% actually have Disease when test is +ive.

Question: 4.167.

Let events:

C_{RR}: selected card has red on both side

C_{RB}: selected card has black on one side

Date: _____ Day: _____

Given:

Total card = 3

Total sides = 4

$P(C_{RR}) = \frac{1}{2} = 0.5$

$P(C_{RB}) = \frac{1}{2} = 0.5$

Let Rup event: the upward face showing on Desk is red.

Assume we Picked: C_{RR}

$P(R_{up} | C_{RR}) = 1$

Assume we Picked: C_{RB}

$P(R_{up} | C_{RB}) = 0.5$

a.

$$P(C_{RR} | R_{up}) = \frac{P(C_{RR}) P(R_{up} | C_{RR})}{P(C_{RR}) P(R_{up} | C_{RR}) + P(C_{RB}) P(R_{up} | C_{RB})}$$

$$P(C_{RR} | R_{up}) = \frac{(0.5)(0.5)}{(0.5)(1) + (0.5)(0.5)}$$

$$P(C_{RR} | R_{up}) = \frac{1}{1.5} = 0.6667$$

$P(C_{RR} | R_{up}) = 66.67\%$.

Date: _____ Day: _____

b-

1. Look at Individual Faces (sides), not cards.
   - C_{RR} has Face 1 (Red) and Face 2 (red)
   - C_{RB} has Face 1 (Red) and Face 2 (Black)
   - There are 4 faced total, 3 of which are red.

2. When we see a Red Face showing.
   Total sides = 4
   Total Red = 3

Card 1:	Card 2:
Side A = Red	Side A = Red
Side B = Red	Side D = black.

we Lift up the hand and seed Red.

other side.

- Possibility 1 (/we are looking at Side A)
  - Flip it → other side is Side B (Red)

- Possibility 2 (we are looking at side B)
  - Flip it → othe side is Side A (Red)

Date: _____ Day: _____

- Possibility 3 (If we are looking at side C)
  - Flip it → other side is side D (Black)

Final:

Out of 3 possible red sides starting up at you.

- In 2 cases other side is Red.
- In 1 case other side is Black.

$P(\text{other side is red}) = 2/3 = 0.6667$

P(other side is red) = 66.67%